

Lab Work: Deep Learning Clustering

Objective

To explore the Deep Learning Clustering by implementing an Unsupervised Deep Autoencoder for feature extraction/clustering and a Supervised Convolutional Neural Network (CNN) for image classification.

Part 1: Deep Learning Clustering (Unsupervised)

Goal: Use a Deep Autoencoder to compress high-dimensional data and cluster the resulting latent representations.

- **Dataset:** MNIST or Fashion-MNIST (ignore labels during training).
- **Architecture:** Construct an **Autoencoder** with an Encoder and Decoder.
 - Encoder: Reduce input dimension (784) down to a latent space of $z \in R^{16}$ or R^{32} .
 - Decoder: Reconstruct the original image from the latent vector z .
- **Clustering Strategy:** After training, extract the latent vectors for the test set. Apply the **K-Means** algorithm to group these vectors into $k = 10$ clusters.
- **Evaluation:** Visualize the latent space using a 2D projection (t-SNE or PCA).

Deliverables & Submission

1. **Source Code:** Clean Jupyter Notebook or Python scripts with documented steps.
2. **Visualizations:** Plots of training/validation loss, t-SNE clustering results, and the confusion matrix.
3. **Comparative Analysis:** A brief discussion (200 words) on why the supervised model outperforms the unsupervised clustering in label accuracy.